DNS Resolver Exposure Assessment Report

Team A3-PS012-TC045 · AIORI Hackathon 3 · PS-012 · ResolverLens 0.2.0 · run **phase4-a** · started 2026-10-26T11:39:32+00:00 · 20 authorised lab servers · 80 probes in 70.8 s

1. Summary

> **8 of 20** servers provide recursion to the external client: t01 (bind), t02 (bind), t03 (unbound), t04 (unbound), t05 (dnsmasq), t14 (dnsmasq), t15 (unbound), t16 (bind). Every decisive verdict matches the seeded ground truth. Accuracy 100.0% (20/20), accuracy on decided cases 100.0%, coverage 80.0% (16 of 20 servers decided; 4 correctly reported as Inconclusive).



Recursive (8) Authoritative-only (4) Refused (4) Inconclusive (4)

How to read the labels. **Recursive**: the server looked up a name for the external client (proved by our own canary server's log). **Authoritative-only**: it answers its own zone but does not recurse. **Refused**: it denies the external client. **Inconclusive**: the evidence is missing or contradictory, and the reason is recorded. Verdicts follow the rules on the method page (section 9).

2. Findings per server

Server	Address	Software	Verdict	Reason	Canary evidence (who asked the canary)	Ground truth (lab only)	Match
t01	10.10.0.11	bind	**Recursive**	-	10.10.0.11	Recursive	**yes**
t02	10.10.0.12	bind	**Recursive**	-	10.10.0.12	Recursive	**yes**
t03	10.10.0.13	unbound	**Recursive**	-	10.10.0.13	Recursive	**yes**
t04	10.10.0.14	unbound	**Recursive**	-	10.10.0.14	Recursive	**yes**
t05	10.10.0.15	dnsmasq	**Recursive**	-	10.10.0.15	Recursive	**yes**
t06	10.10.0.16	bind	**Refused**	-	-	Refused	**yes**
t07	10.10.0.17	bind	**Refused**	-	-	Refused	**yes**
t08	10.10.0.18	unbound	**Refused**	-	-	Refused	**yes**
t09	10.10.0.19	unbound	**Refused**	-	-	Refused	**yes**
t10	10.10.0.20	bind	**Authoritative-only**	-	-	Authoritative-only	**yes**
t11	10.10.0.21	bind	**Authoritative-only**	-	-	Authoritative-only	**yes**
t12	10.10.0.22	nsd	**Authoritative-only**	-	-	Authoritative-only	**yes**
t13	10.10.0.23	nsd	**Authoritative-only**	-	-	Authoritative-only	**yes**
t14	10.10.0.24	dnsmasq	**Recursive**	-	10.10.0.11	Recursive	**yes**
t15	10.10.0.25	unbound	**Recursive**	-	10.10.0.13	Recursive	**yes**
t16	10.10.0.26	bind	**Recursive**	-	10.10.0.15	Recursive	**yes**
t17	10.10.0.27	unbound	**Inconclusive**	TIMEOUT	-	Inconclusive	**yes**
t18	10.10.0.28	bind	**Inconclusive**	TIMEOUT	-	Inconclusive	**yes**
t19	10.10.0.29	dnsmasq	**Inconclusive**	TIMEOUT	-	Inconclusive	**yes**
t20	10.10.0.30	bind	**Inconclusive**	TIMEOUT	-	Inconclusive	**yes**

The ground-truth column exists only because this is a seeded lab. A real assessment has no such column.

3. Evaluation against the seeded ground truth

Rows are the expected labels, columns are what ResolverLens reported. Green cells are correct, red cells are errors.

Expected \ Reported	Recursive	Authoritative-only	Refused	Inconclusive	Total
Recursive	8	0	0	0	8
Authoritative-only	0	4	0	0	4
Refused	0	0	4	0	4
Inconclusive	0	0	0	4	4

Class	Precision	Recall	F1	Expected	Reported
Recursive	1.00	1.00	1.00	8	8
Authoritative-only	1.00	1.00	1.00	4	4
Refused	1.00	1.00	1.00	4	4
Inconclusive	1.00	1.00	1.00	4	4

Measure	Value	Meaning
Accuracy	100.0%	Verdicts equal to the expected label, over all servers
Accuracy on decided cases	100.0%	Correct verdicts among servers where the tool gave a decisive label
Coverage	80.0%	Share of servers the tool decided instead of reporting Inconclusive
Wrong decisions	0	Decisive verdicts that disagree with the expected label (the costly error)
False exposure / missed exposure	0 / 0	Reported Recursive when it should not be / failed to report Recursive when it should
Correct abstentions	4	Inconclusive reported for servers seeded as unusable

4. False and inconclusive result analysis

4.1 False results

> No false results in this run: all 20 verdicts equal their expected labels. Each false result would be listed here with a likely cause (ACL, caching, forwarder chain, packet loss, rate limiting, upstream failure).

4.2 Inconclusive results

Server	Software	Reason	Likely cause	Why the tool did not guess	Seeded as (lab only)	Expected?
t17	unbound	TIMEOUT	packet loss, rate limiting or filtering	No reply to any canary attempt. A silent drop looks the same as packet loss, so the tool does not guess.	slow	**yes**
t18	bind	TIMEOUT	packet loss, rate limiting or filtering	No reply to any canary attempt. A silent drop looks the same as packet loss, so the tool does not guess.	rate limited	**yes**
t19	dnsmasq	TIMEOUT	packet loss, rate limiting or filtering	No reply to any canary attempt. A silent drop looks the same as packet loss, so the tool does not guess.	silent	**yes**
t20	bind	TIMEOUT	packet loss, rate limiting or filtering	No reply to any canary attempt. A silent drop looks the same as packet loss, so the tool does not guess.	broken upstream	**yes**

5. Exposure and remediation

RFC 5358 asks operators to offer recursion only to their intended clients. The following servers answered an external client:

Server	Software	Suggested change (check the syntax for your version)
t01 (10.10.0.11)	bind	`allow-recursion { <your clients>; localhost; }; allow-query-cache { <your clients>; localhost; };`
t02 (10.10.0.12)	bind	`allow-recursion { <your clients>; localhost; }; allow-query-cache { <your clients>; localhost; };`
t03 (10.10.0.13)	unbound	`access-control: 0.0.0.0/0 refuse then access-control: <your clients> allow`
t04 (10.10.0.14)	unbound	`access-control: 0.0.0.0/0 refuse then access-control: <your clients> allow`
t05 (10.10.0.15)	dnsmasq	`local-service (or interface=<internal interface> with bind-interfaces)`
t14 (10.10.0.24)	dnsmasq	`local-service (or interface=<internal interface> with bind-interfaces)` Its lookups came from 10.10.0.11: that upstream server needs the same restriction.
t15 (10.10.0.25)	unbound	`access-control: 0.0.0.0/0 refuse then access-control: <your clients> allow` Its lookups came from 10.10.0.13: that upstream server needs the same restriction.
t16 (10.10.0.26)	bind	`allow-recursion { <your clients>; localhost; }; allow-query-cache { <your clients>; localhost; };` Its lookups came from 10.10.0.15: that upstream server needs the same restriction.

6. Safety and compliance, measured from the evidence

Check	Observed	Limit or rule
Destinations contacted	20 (all on the allowlist: yes)	at most 20 authorised lab servers
Query types sent	A	type A only, never ANY
Peak rate, all targets together	3 queries in any second	5.0 per second plus burst
Peak rate, one target	3 queries in any second	1.0 per second plus burst
Largest reply / reply-to-query ratio	251 bytes / 6.28	1232 bytes / 10.0
How the run ended	completed	safety tripwire and stop file available

The tool has no spoofing option: every query leaves from the probe client's own address. The lab network has no Internet access.

7. Evidence per server

t01 · 10.10.0.11 · bind · Recursive

3/3 canary attempts: NOERROR, RA, marker answer, canary log hit (rule R1)

Query	RD	Status	RA	AA	Answer	Canary log	ms
baseline : ns.lab.test	0	NOERROR	1	0	-	-	13.4
canary 1: rl-a5920c1a57.canary.lab	1	NOERROR	1	0	192.0.2.53	seen from 10.10.0.11	8.2
canary 2: rl-356890815c.canary.lab	1	NOERROR	1	0	192.0.2.53	seen from 10.10.0.11	2.1
canary 3: rl-67ce440595.canary.lab	1	NOERROR	1	0	192.0.2.53	seen from 10.10.0.11	4.7

t02 · 10.10.0.12 · bind · Recursive

3/3 canary attempts: NOERROR, RA, marker answer, canary log hit (rule R1)

Query	RD	Status	RA	AA	Answer	Canary log	ms
baseline : ns.lab.test	0	NOERROR	1	0	-	-	6.4
canary 1: rl-cac1021e6c.canary.lab	1	NOERROR	1	0	192.0.2.53	seen from 10.10.0.12	5.0
canary 2: rl-df48d9212a.canary.lab	1	NOERROR	1	0	192.0.2.53	seen from 10.10.0.12	3.3
canary 3: rl-61c62edeef.canary.lab	1	NOERROR	1	0	192.0.2.53	seen from 10.10.0.12	2.7

t03 · 10.10.0.13 · unbound · Recursive

3/3 canary attempts: NOERROR, RA, marker answer, canary log hit (rule R1)

Query	RD	Status	RA	AA	Answer	Canary log	ms
baseline : ns.lab.test	0	NXDOMAIN	1	1	-	-	2.4
canary 1: rl-ed8733bf6d.canary.lab	1	NOERROR	1	0	192.0.2.53	seen from 10.10.0.13	2.1
canary 2: rl-5b8798b5a2.canary.lab	1	NOERROR	1	0	192.0.2.53	seen from 10.10.0.13	2.5
canary 3: rl-5d7e1d4e50.canary.lab	1	NOERROR	1	0	192.0.2.53	seen from 10.10.0.13	2.0

t04 · 10.10.0.14 · unbound · Recursive

3/3 canary attempts: NOERROR, RA, marker answer, canary log hit (rule R1)

Query	RD	Status	RA	AA	Answer	Canary log	ms
baseline : ns.lab.test	0	NXDOMAIN	1	1	-	-	1.6
canary 1: rl-06eaba7c6a.canary.lab	1	NOERROR	1	0	192.0.2.53	seen from 10.10.0.14	1.8
canary 2: rl-17d43e6e9a.canary.lab	1	NOERROR	1	0	192.0.2.53	seen from 10.10.0.14	4.5
canary 3: rl-258c213b26.canary.lab	1	NOERROR	1	0	192.0.2.53	seen from 10.10.0.14	2.1

t05 · 10.10.0.15 · dnsmasq · Recursive

3/3 canary attempts: NOERROR, RA, marker answer, canary log hit (rule R1)

Query	RD	Status	RA	AA	Answer	Canary log	ms
baseline : ns.lab.test	0	REFUSED	1	0	-	-	1.6
canary 1: rl-acbc9ffe86.canary.lab	1	NOERROR	1	1	192.0.2.53	seen from 10.10.0.15	2.2
canary 2: rl-6ca65a8687.canary.lab	1	NOERROR	1	1	192.0.2.53	seen from 10.10.0.15	1.7
canary 3: rl-4ab4e21376.canary.lab	1	NOERROR	1	1	192.0.2.53	seen from 10.10.0.15	1.9

t06 · 10.10.0.16 · bind · Refused

REFUSED for the baseline and 3/3 canary attempts (rule R2)

Query	RD	Status	RA	AA	Answer	Canary log	ms
baseline : ns.lab.test	0	REFUSED	0	0	-	-	2.5
canary 1: rl-3d5df599fb.canary.lab	1	REFUSED	0	0	-	-	1.7
canary 2: rl-6e2d55a5d4.canary.lab	1	REFUSED	0	0	-	-	1.0
canary 3: rl-e7e45988d0.canary.lab	1	REFUSED	0	0	-	-	1.0

t07 · 10.10.0.17 · bind · Refused

REFUSED for the baseline and 3/3 canary attempts (rule R2)

Query	RD	Status	RA	AA	Answer	Canary log	ms
baseline : ns.lab.test	0	REFUSED	0	0	-	-	1.9
canary 1: rl-7d11f9b8c1.canary.lab	1	REFUSED	0	0	-	-	2.7
canary 2: rl-cc1f5fa6b5.canary.lab	1	REFUSED	0	0	-	-	1.3
canary 3: rl-c24965aec3.canary.lab	1	REFUSED	0	0	-	-	1.6

t08 · 10.10.0.18 · unbound · Refused

REFUSED for the baseline and 3/3 canary attempts (rule R2)

Query	RD	Status	RA	AA	Answer	Canary log	ms
baseline : ns.lab.test	0	REFUSED	0	0	-	-	0.9
canary 1: rl-1729c2ec4a.canary.lab	1	REFUSED	0	0	-	-	0.7
canary 2: rl-a6d31b6970.canary.lab	1	REFUSED	0	0	-	-	0.8
canary 3: rl-95fbd19e9d.canary.lab	1	REFUSED	0	0	-	-	0.9

t09 · 10.10.0.19 · unbound · Refused

REFUSED for the baseline and 3/3 canary attempts (rule R2)

Query	RD	Status	RA	AA	Answer	Canary log	ms
baseline : ns.lab.test	0	REFUSED	0	0	-	-	0.9
canary 1: rl-a0871b0ea6.canary.lab	1	REFUSED	0	0	-	-	0.8
canary 2: rl-a1af614d64.canary.lab	1	REFUSED	0	0	-	-	0.8
canary 3: rl-ecae9924b8.canary.lab	1	REFUSED	0	0	-	-	0.9

t10 · 10.10.0.20 · bind · Authoritative-only

baseline answered with AA; 3/3 canary attempts got no answer and no canary hit (rule R3)

Query	RD	Status	RA	AA	Answer	Canary log	ms
baseline : ns.lab.test	0	NOERROR	0	1	10.10.0.20	-	2.0
canary 1: rl-0c2a1c804f.canary.lab	1	REFUSED	0	0	-	-	1.1
canary 2: rl-bbd1865f63.canary.lab	1	REFUSED	0	0	-	-	3.1
canary 3: rl-84045dec41.canary.lab	1	REFUSED	0	0	-	-	1.1

t11 · 10.10.0.21 · bind · Authoritative-only

baseline answered with AA; 3/3 canary attempts got no answer and no canary hit (rule R3)

Query	RD	Status	RA	AA	Answer	Canary log	ms
baseline : ns.lab.test	0	NOERROR	0	1	10.10.0.21	-	2.1
canary 1: rl-4bc01ea058.canary.lab	1	REFUSED	0	0	-	-	1.1
canary 2: rl-658ad9acbf.canary.lab	1	REFUSED	0	0	-	-	1.4
canary 3: rl-f491977ca8.canary.lab	1	REFUSED	0	0	-	-	1.4

t12 · 10.10.0.22 · nsd · Authoritative-only

baseline answered with AA; 3/3 canary attempts got no answer and no canary hit (rule R3)

Query	RD	Status	RA	AA	Answer	Canary log	ms
baseline : ns.lab.test	0	NOERROR	0	1	10.10.0.22	-	1.8
canary 1: rl-8cc55ae37f.canary.lab	1	REFUSED	0	0	-	-	1.1
canary 2: rl-e4d1449394.canary.lab	1	REFUSED	0	0	-	-	1.5
canary 3: rl-3f684df3bc.canary.lab	1	REFUSED	0	0	-	-	3.0

t13 · 10.10.0.23 · nsd · Authoritative-only

baseline answered with AA; 3/3 canary attempts got no answer and no canary hit (rule R3)

Query	RD	Status	RA	AA	Answer	Canary log	ms
baseline : ns.lab.test	0	NOERROR	0	1	10.10.0.23	-	2.2
canary 1: rl-cfb65a1b8f.canary.lab	1	REFUSED	0	0	-	-	2.1
canary 2: rl-ec5d174fde.canary.lab	1	REFUSED	0	0	-	-	1.1
canary 3: rl-0931e936a6.canary.lab	1	REFUSED	0	0	-	-	1.2

t14 · 10.10.0.24 · dnsmasq · Recursive

3/3 canary attempts: NOERROR, RA, marker answer, canary log hit (rule R1)

Query	RD	Status	RA	AA	Answer	Canary log	ms
baseline : ns.lab.test	0	NOERROR	1	0	-	-	6.3
canary 1: rl-446174e04a.canary.lab	1	NOERROR	1	0	192.0.2.53	seen from 10.10.0.11	3.3
canary 2: rl-633aee4c69.canary.lab	1	NOERROR	1	0	192.0.2.53	seen from 10.10.0.11	5.2
canary 3: rl-85e8be92fb.canary.lab	1	NOERROR	1	0	192.0.2.53	seen from 10.10.0.11	3.2

t15 · 10.10.0.25 · unbound · Recursive

3/3 canary attempts: NOERROR, RA, marker answer, canary log hit (rule R1)

Query	RD	Status	RA	AA	Answer	Canary log	ms
baseline : ns.lab.test	0	NXDOMAIN	1	1	-	-	1.8
canary 1: rl-e6684ce523.canary.lab	1	NOERROR	1	0	192.0.2.53	seen from 10.10.0.13	2.5
canary 2: rl-fa77a2c0bf.canary.lab	1	NOERROR	1	0	192.0.2.53	seen from 10.10.0.13	2.2
canary 3: rl-0879dac8f1.canary.lab	1	NOERROR	1	0	192.0.2.53	seen from 10.10.0.13	2.7

t16 · 10.10.0.26 · bind · Recursive

3/3 canary attempts: NOERROR, RA, marker answer, canary log hit (rule R1)

Query	RD	Status	RA	AA	Answer	Canary log	ms
baseline : ns.lab.test	0	NOERROR	1	0	-	-	4.7
canary 1: rl-596d39d2e8.canary.lab	1	NOERROR	1	0	192.0.2.53	seen from 10.10.0.15	3.1
canary 2: rl-dc85904b62.canary.lab	1	NOERROR	1	0	192.0.2.53	seen from 10.10.0.15	3.8
canary 3: rl-339e7d3eef.canary.lab	1	NOERROR	1	0	192.0.2.53	seen from 10.10.0.15	2.9

t17 · 10.10.0.27 · unbound · Inconclusive (TIMEOUT)

no reply to any of the 3 canary attempts (rule R4)

Query	RD	Status	RA	AA	Answer	Canary log	ms
baseline : ns.lab.test	0	TIMEOUT	0	0	-	-	-
canary 1: rl-40c3f5037d.canary.lab	1	TIMEOUT	0	0	-	-	-
canary 2: rl-f469196b4c.canary.lab	1	TIMEOUT	0	0	-	-	-
canary 3: rl-1a9675a23f.canary.lab	1	TIMEOUT	0	0	-	-	-

t18 · 10.10.0.28 · bind · Inconclusive (TIMEOUT)

no reply to any of the 3 canary attempts (rule R4)

Query	RD	Status	RA	AA	Answer	Canary log	ms
baseline : ns.lab.test	0	NOERROR	1	0	-	-	5.7
canary 1: rl-4f67414e5b.canary.lab	1	TIMEOUT	0	0	-	-	-
canary 2: rl-06c4c84d20.canary.lab	1	TIMEOUT	0	0	-	-	-
canary 3: rl-e176593efc.canary.lab	1	TIMEOUT	0	0	-	-	-

t19 · 10.10.0.29 · dnsmasq · Inconclusive (TIMEOUT)

no reply to any of the 3 canary attempts (rule R4)

Query	RD	Status	RA	AA	Answer	Canary log	ms
baseline : ns.lab.test	0	TIMEOUT	0	0	-	-	-
canary 1: rl-585276b21a.canary.lab	1	TIMEOUT	0	0	-	-	-
canary 2: rl-c92cc9d79e.canary.lab	1	TIMEOUT	0	0	-	-	-
canary 3: rl-73260a5450.canary.lab	1	TIMEOUT	0	0	-	-	-

t20 · 10.10.0.30 · bind · Inconclusive (TIMEOUT)

no reply to any of the 3 canary attempts (rule R4)

Query	RD	Status	RA	AA	Answer	Canary log	ms
baseline : ns.lab.test	0	NOERROR	1	0	-	-	6.2
canary 1: rl-427474cc37.canary.lab	1	TIMEOUT	0	0	-	-	-
canary 2: rl-48e4be8280.canary.lab	1	TIMEOUT	0	0	-	-	-
canary 3: rl-4cc623c679.canary.lab	1	TIMEOUT	0	0	-	-	-

8. Reproducibility

Item	Value
Run, seed, tool version	phase4-a, 1337, 0.2.0
Config digest (settings and targets file)	`a79e23245d905be41ed801a58b2f76ea343f1974bbfe7dcc4078dfdcb49b0a54`
Timeout, attempts, EDNS buffer	2.0 s, 3, 1232 bytes
Re-run	`make probe RUN=<name>` then `make report RUN=<name>`. The same seed and run id give the same canary names, and the verdicts are computed from evidence.jsonl alone.

9. Method: the four rules

Each server receives one baseline query (RD off, a name in its own zone) and 3 canary queries (RD on, a fresh unique name each time, in a zone served only by our canary server). The canary server logs every query it receives. Rules are checked in order and the first match decides:

Rule	Verdict	Condition
1	Recursive	Every canary attempt is NOERROR with RA set, the marker answer, and a hit in the canary log
2	Refused	The baseline and every canary attempt are REFUSED
3	Authoritative-only	The baseline is answered with AA; every canary attempt gets a reply with no answer, no RA and no canary hit
4	Inconclusive	Anything else, with a reason: TIMEOUT, SERVFAIL, TC_FAIL, CONFLICT, ERROR or NO_DATA